

Title

Modelling Climate-Driven Ambystoma tigrinum Virus: Integrating Temperature, Landscape Fragmentation, and Host Behaviour

Authors and Affiliations

Andrew Kerfoot

NAU School of Informatics, Computing, and Cyber Systems

Abstract

Ambystoma tigrinum virus (ATV) is a ranavirus associated with severe population declines in tiger salamanders (*Ambystoma tigrinum*), yet the ecological processes driving variation in outbreak timing and severity remain poorly understood. In Arizona, salamander populations occur as spatially separated pond networks, where host movement, habitat fragmentation, and environmental variability may strongly influence disease dynamics. To investigate these interactions, I developed a spatially explicit compartmental disease model in Python using the epymorph framework. The model represented multiple connected ponds with local within-pond transmission, seasonal adult movement between breeding habitats, and temperature-dependent transmission using daily climate data. A stepwise set of scenarios compared baseline within-pond dynamics, the effects of host movement, and the addition of environmental forcing. Results showed that local transmission alone produced relatively isolated outbreaks, while seasonal movement increased spread among ponds and altered epidemic timing. Incorporating temperature further changed outbreak magnitude and seasonal synchrony, indicating that disease outcomes depend on the overlap of host phenology, connectivity, and

favourable environmental conditions. These findings suggest that ATV dynamics are best understood as metapopulation processes shaped by multiple interacting drivers rather than isolated local events. More broadly, this study demonstrates how spatially explicit mechanistic models can be used to evaluate wildlife disease risk under future climate and landscape change.

Introduction

Arizona tiger salamander (*Ambystoma tigrinum*) populations across the American Southwest are experiencing severe disease-driven declines associated with *Ambystoma tigrinum* virus (ATV), a ranavirus capable of causing mass mortality events that can eliminate entire pond populations within a single season (Altizer et al., 2006; Brunner & Yarber, 2018). ATV spreads through both direct host contact and viral shedding into the aquatic environment, and outbreaks often disproportionately affect larval life stages, with seasonal mortality exceeding 90% in some epizootic events (Terrell et al., 2013). Despite the severity of these outbreaks, the mechanisms determining whether a local population recovers, persists, or collapses following exposure remain poorly understood. This uncertainty has important conservation implications in a warming and increasingly fragmented landscape.

A major challenge is that tiger salamander populations do not function as isolated units. Across Arizona, salamanders occupy spatially separated ponds, stock tanks, and seasonal wetlands that vary in hydroperiod, elevation, and landscape connectivity. Adults may disperse between breeding ponds during late winter and early spring, while larvae remain restricted to aquatic natal habitats. As a result, local populations function as a network of connected subpopulations rather than as independent sites (Figure 1). This spatial structure means that

outbreak dynamics are shaped not only by within-pond transmission, but also by host movement and potential pathogen transport across the landscape (Terrell et al., 2013).

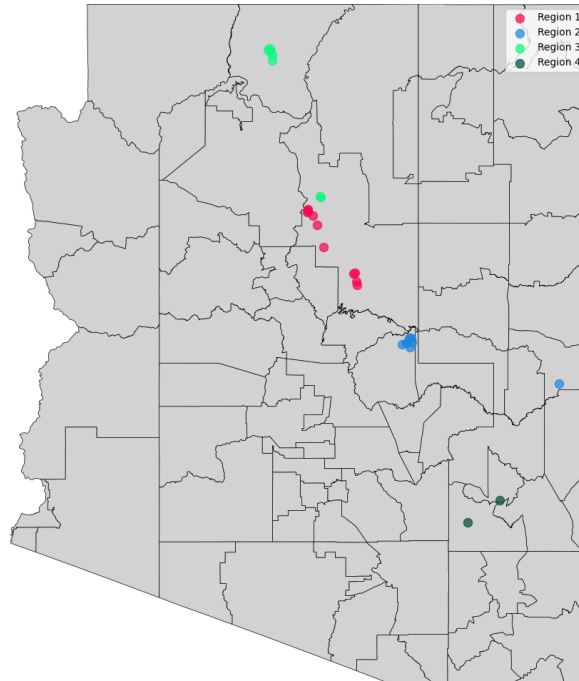


Figure 1. Spatial distribution of ATV-positive sampling sites across Arizona, grouped into four broad geographic regions. Points represent pond sites with historical ATV-positive records, and colours indicate regional groupings used to summarize spatial variation in observed infection patterns.

Environmental persistence of ATV further complicates this system. Ranaviruses may remain infectious in water or sediment after host shedding, allowing infection timing to become partially decoupled from direct contact and creating the potential for delayed or spatially extended outbreaks (Brunner & Yarber, 2018). Temperature adds another important layer of complexity: warmer conditions can increase viral replication, suppress amphibian immune

responses, and alter the timing or severity of epizootics (Rojas et al., 2005; Rollins-Smith & Le Sage, 2023; Terrell et al., 2013).

These interacting drivers are important because similar processes shape many wildlife disease systems. Host movement can spread pathogens among otherwise separated populations, while habitat fragmentation can alter how often populations interact, recover, or become recolonized after local declines. Climate change may further shift disease risk by changing the timing of host reproduction, pathogen replication, immune performance, and seasonal transmission windows. In amphibians and other ectotherms, these effects may be especially strong because body temperature, development, immune function, and pathogen growth are closely tied to environmental conditions. For ATV, this means that outbreak risk is unlikely to depend on pathogen presence alone; instead, severe outbreaks may depend on whether susceptible larvae, infected or carrier adults, connected ponds, and favourable temperatures overlap in time and space.

Previous research has documented ATV prevalence patterns at individual ponds, described experimental temperature effects, and examined ranavirus transmission within localized systems. However, these efforts have largely treated outbreaks in isolation and have not fully integrated movement among breeding sites, demographic timing, and environmental forcing into a unified spatial framework. A mechanistic model that combines these processes could help explain observed heterogeneity in outbreak timing and severity while improving predictions under future climate and landscape change.

To address this gap, this study develops a spatially explicit compartmental disease model using the epymorph framework to investigate how within-pond transmission, seasonal host

movement, habitat connectivity, and climate variability interact to shape ATV dynamics across Arizona tiger salamander populations.

Methods

Modelling Framework

To investigate these processes, I developed a spatially explicit compartmental disease model in Python 3.12 using the epymorph framework (version 1.1.0). The model was designed to represent multiple connected breeding ponds, with disease dynamics occurring both within ponds and across the landscape through seasonal host movement. Each pond was treated as a node in a metapopulation network, where local transmission occurred within nodes and dispersal linked nodes through time-varying movement processes. Model development followed an iterative structure. Simpler versions were first used to verify demographic transitions and disease behaviour before adding movement and environmental forcing. This stepwise approach allowed the effects of each added mechanism to be evaluated separately.

Within-Pond Disease Dynamics

Within each pond, salamanders were divided into biologically relevant disease and life-stage compartments. The current model structure included susceptible offspring or juveniles (S_off), infected offspring or juveniles (I_off), recovered or noninfectious adults (R_a), and chronic or asymptomatic carrier adults (R_c). Susceptible offspring could become infected through contact with infectious individuals or environmentally mediated exposure. Infected individuals could die from the disease, survive the infection and mature into adults, or transition into a chronic carrier state. Adults were treated as the primary dispersing life stage, whereas larval stages remained within natal ponds. This reflects metamorphosis in amphibians, where

larvae develop in aquatic breeding ponds and generally become capable of overland dispersal only after transitioning into later terrestrial or semi-terrestrial stages. Natural mortality and seasonal recruitment were also included. A seasonal birth pulse occurred during the breeding period based on the number of adults present in each pond at that time. This structure allowed local demography and disease transmission to interact through time.

Between-Pond Movement

To represent breeding migrations, adults were allowed to move between ponds during defined seasonal movement windows. Movement probabilities were structured using inter-pond distance, such that nearby ponds were more likely to exchange individuals than distant ponds. Adults were assumed to disperse into breeding habitats before or during reproduction, remain temporarily, and later return or redistribute following breeding. Because infected or carrier adults may transport ATV between sites, movement created a mechanism for pathogen introduction into previously uninfected ponds. Movement timing was modelled as a seasonal process rather than as constant dispersal, since overlap among adult arrival, larval presence, and favourable transmission conditions was expected to strongly influence outbreak initiation.

Environmental Forcing

Later model versions incorporated daily environmental temperature as a driver of transmission dynamics. Temperature data were obtained from PRISM daily climate products and linked to simulation days. Transmission parameters were allowed to vary through time as a function of temperature rather than remaining constant. This approach was motivated by previous work showing that ranavirus replication, host susceptibility, and disease severity may be temperature dependent (Rojas et al., 2005; Rollins-Smith & Le Sage, 2023). Under this

framework, outbreaks were expected to emerge when host presence, breeding phenology, and favourable transmission temperatures overlapped.

Conceptual Seasonal Framework

Model development was guided by the hypothesis that ATV outbreaks occur when host movement, breeding phenology, and environmentally favourable transmission conditions overlap seasonally (Figure 2). In this framework, adult salamanders disperse to breeding ponds before reproduction, larvae are produced during the breeding window, and outbreaks emerge when susceptible larvae are present during periods of elevated transmission potential. This conceptual structure was used to organize the timing of adult movement, birth pulses, maturation, and temperature-dependent transmission in the model. By separating these processes conceptually, each model scenario could test how adding movement and environmental forcing changed outbreak timing, magnitude, and spread across ponds.

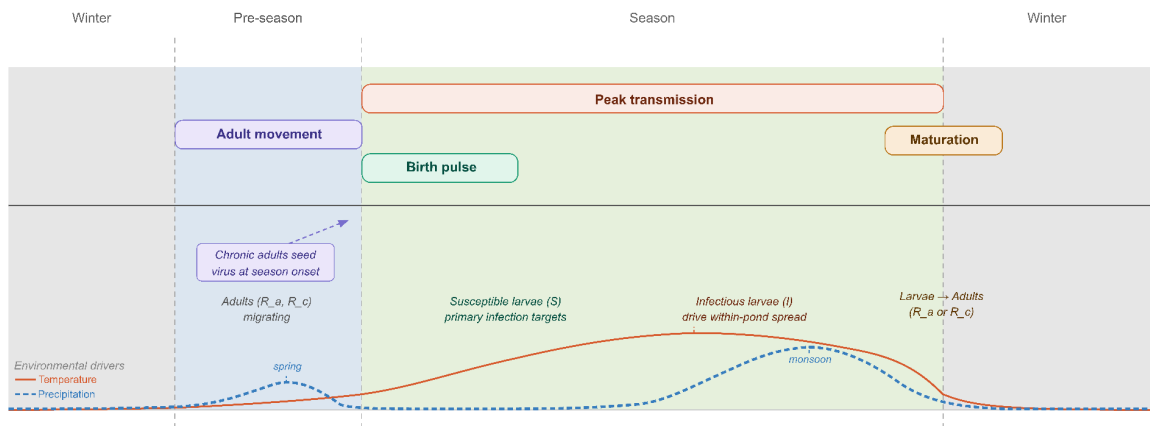


Figure 2. Conceptual seasonal framework linking host movement, breeding phenology, and transmission opportunity. Adults are assumed to move into breeding habitats during the pre-season, larvae are produced during a seasonal birth pulse, and transmission risk increases

when susceptible larvae are present during periods of favourable environmental conditions. The diagram illustrates the central hypothesis that outbreaks are most likely when movement, larval presence, and transmission potential overlap in time.

Model Scenarios

To evaluate the relative importance of movement and environmental forcing, a series of increasingly complex model scenarios was developed (Table 1). Each scenario added one major ecological process to the previous model, allowing changes in outbreak timing, outbreak magnitude, and synchrony among ponds to be compared across model structures.

Table 1. Stepwise model scenarios used to evaluate ATV dynamics. Each scenario adds ecological complexity to the previous model, allowing the effects of local transmission, seasonal movement, and environmental forcing to be evaluated separately.

Model scenario	Processes included	Purpose
Model 1: Baseline	Within-pond disease dynamics only; no movement or environmental forcing	Establishes baseline local ATV dynamics and shows how outbreaks behave without spatial spread or climate effects
Model 2: Seasonal movement	Within-pond disease dynamics plus seasonal adult movement among ponds	Tests whether adult movement can spread ATV among connected ponds and alter outbreak timing
Model 3: Temperature and movement	Seasonal movement plus temperature-dependent transmission	Tests whether environmental forcing changes outbreak magnitude, timing, and synchrony when transmission varies through the season
Model 4: Regional climate implementation	Pond-specific environmental conditions and broader regional temperature heterogeneity	Future model extension designed to compare outbreak dynamics across northern, central, and southern Arizona climates

This stepwise structure made it possible to interpret how each added mechanism changed simulated ATV dynamics across connected pond populations.

Observed ATV Pattern Data

Observed ATV patterns were summarized using historical ATV sampling records from surveyed pond sites in Arizona. Site locations were grouped into four broad geographic regions to visualize the spatial distribution of ATV-positive records across the state. For temporal summaries, sampling dates were used to calculate the cumulative proportion of infected observations through each season, both across all surveyed ponds and separately by geographic region. These observed patterns were not used to calibrate model parameters directly; instead, they provided empirical motivation for evaluating whether spatial connectivity, seasonal timing, and environmental forcing could help explain variation in outbreak timing and severity.

Model Assumptions

The model makes several simplifying assumptions intended to represent major biological processes while keeping the system interpretable. First, each pond is treated as a discrete habitat node, and individuals within each pond are assumed to mix homogeneously. Second, movement is restricted to adults because tiger salamander larvae remain aquatic until metamorphosis and are not expected to disperse among ponds before transitioning into terrestrial or semi-terrestrial adult stages. Third, adult movement occurs during a defined seasonal window associated with breeding migration rather than continuously throughout the year. Fourth, births occur as a seasonal pulse based on the number of adults present at breeding sites, reflecting the seasonal reproductive timing of tiger salamanders. Fifth, infected offspring may die, survive infection, or mature into adult classes, while disease-related mortality removes individuals from the simulated

population rather than tracking carcasses or environmental viral material separately. Sixth, the model does not include long-term protective immunity; instead, adults are represented as recovered or noninfectious adults and chronic carrier adults, allowing persistent infection risk without adding additional immune-history classes. Finally, temperature modifies transmission potential, but the model does not separately represent every physiological mechanism linking temperature, immunity, viral replication, mortality, and host behaviour. These assumptions are biologically motivated but simplify the true system, and the most limiting assumptions are revisited in the Discussion.

Simulation Output and Visualization

For each scenario, compartment abundances were tracked through time for each pond. Primary outputs included timing of infection emergence, peak outbreak size, persistence of infection, and differences among connected populations. Results were visualized using time-series plots and comparative scenario figures. These outputs were used to assess how connectivity, seasonal movement, and climate forcing alter ATV transmission dynamics across Arizona tiger salamander metapopulations.

Results

Results are presented first for observational patterns motivating the model, followed by outputs from progressively more complex simulation scenarios.

Observed Spatial and Temporal Patterns

Historical ATV-positive sites were distributed across multiple regions of Arizona, indicating that infections occur across ecologically distinct parts of the state (Figure 1). This

broad regional distribution supports the use of a spatial modelling framework rather than treating outbreaks as isolated local events.

Cumulative infection prevalence differed substantially between years. In 2019, infections increased earlier in the season, rose more rapidly, and reached a higher cumulative proportion of infected than in 2020 (Figure 3). In contrast, 2020 exhibited a slower rise and lower final prevalence. These differences suggest that annual environmental conditions or host dynamics may strongly influence outbreak severity.

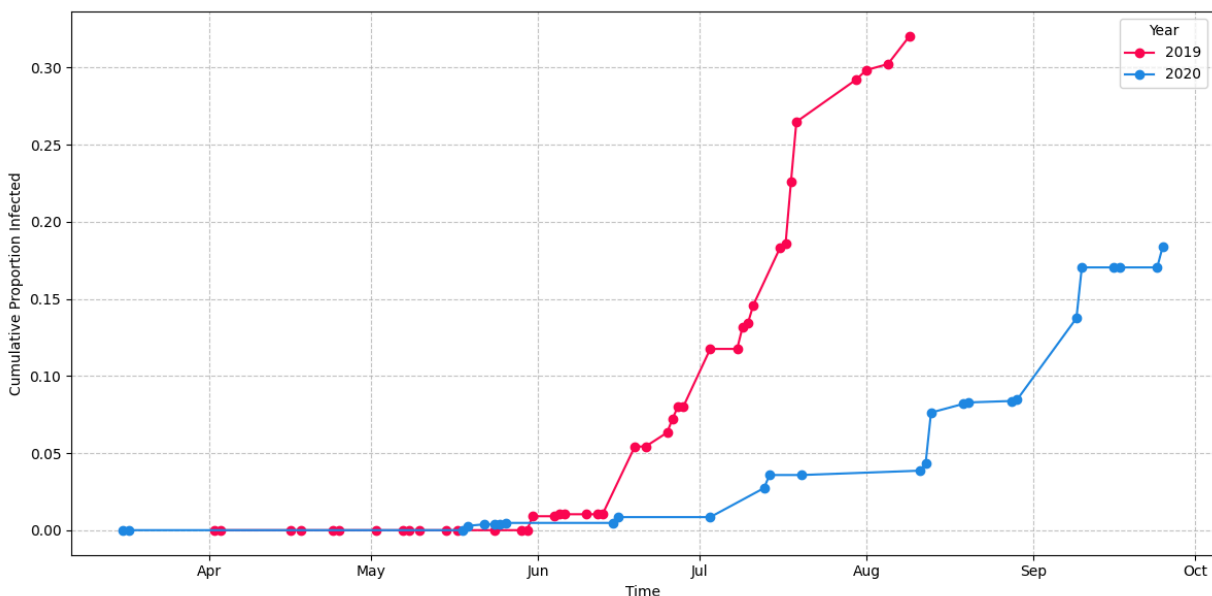


Figure 3. Cumulative proportion infected with ATV by year. The steeper increase in 2019 indicates earlier and more rapid accumulation of positive observations, while the slower increase in 2020 suggests lower or later seasonal infection prevalence.

When separated by region, prevalence patterns also varied spatially (Figure 4). In 2019, northern regions reached higher cumulative prevalence than southern regions, whereas in 2020 this pattern shifted, with the Mogollon Rim region exhibiting comparatively greater infection

levels. Together, these results indicate that ATV dynamics vary across both time and space, motivating the need for a mechanistic model that incorporates movement and environmental forcing.

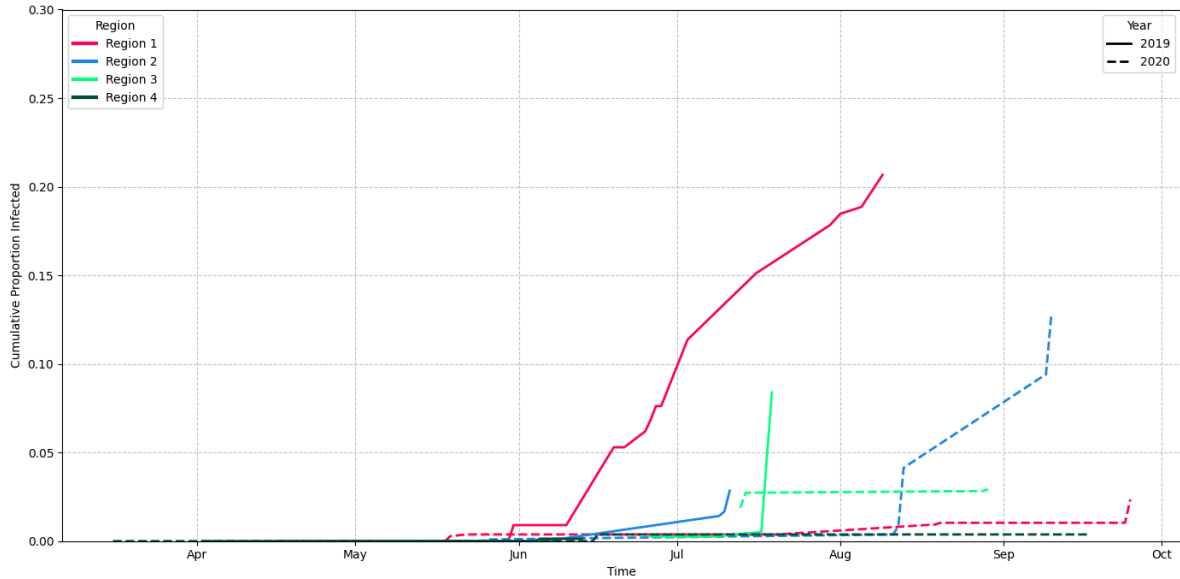


Figure 4. Cumulative proportion infected with ATV by year and region. Lines show regional differences in the timing and magnitude of observed ATV prevalence, demonstrating that infection patterns varied across both space and year.

Model 1: Baseline

Under the baseline scenario with no movement and no environmental forcing, outbreaks remained localized within ponds and followed similar trajectories across the system (Figure 5). Infection timing and magnitude were primarily determined by initial conditions and within-pond transmission parameters. No mechanism existed for spread between ponds or seasonal synchronization.

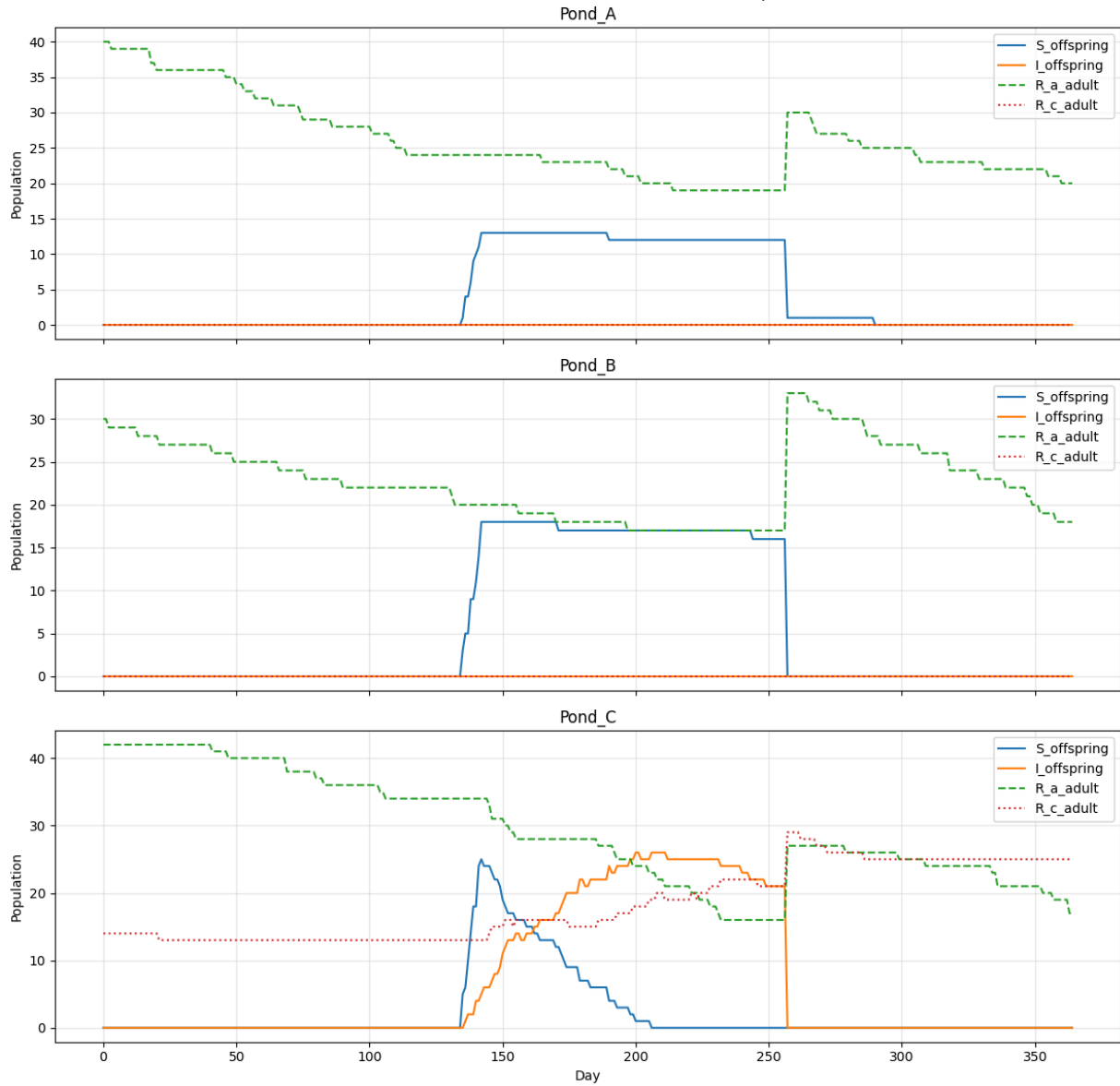


Figure 5. Model 1 output: within-pond ATV dynamics without movement or environmental forcing. Lines show simulated abundances of susceptible offspring, infected offspring, recovered/noninfectious adults, and chronic carrier adults across three ponds. Without movement or temperature-dependent transmission, outbreaks remain localized and are driven primarily by initial conditions and within-pond transmission.

Model 2: Seasonal Movement

When seasonal adult movement was added, infection dynamics changed substantially (Figure 6). Movement allowed infected or carrier adults to introduce ATV into previously uninfected ponds, creating secondary outbreaks and increasing synchrony among connected sites. Outbreak timing became dependent on the overlap between movement windows and host recruitment periods.

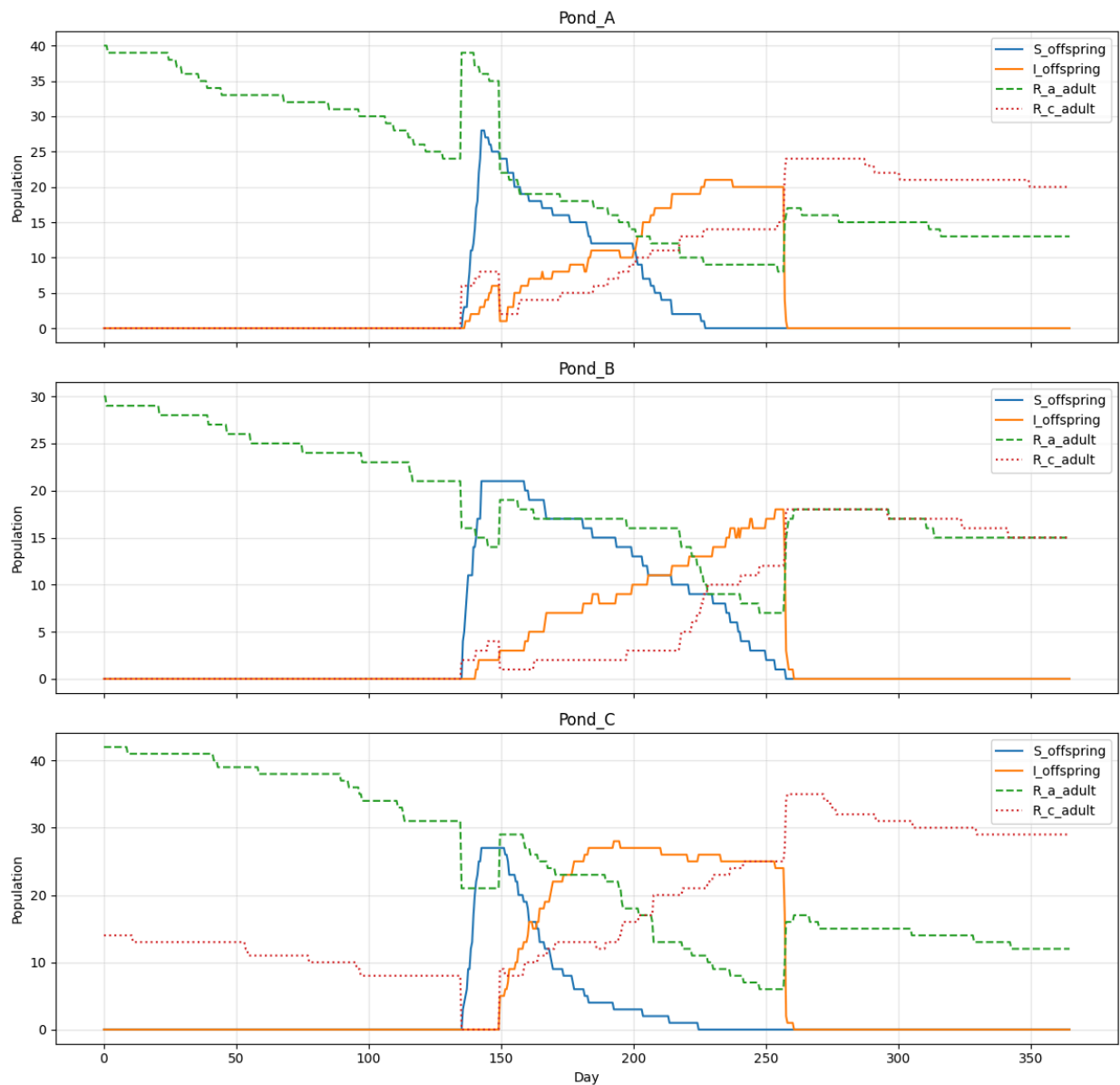


Figure 6. Model 2 output: effects of seasonal adult movement on ATV dynamics. Lines show simulated compartment abundances through time after adding adult movement among ponds during the breeding season. Movement allows infected or chronic carrier adults to introduce ATV into additional ponds, creating secondary outbreaks and changing the timing of infection across the system.

Model 3: Temperature and Movement

Incorporating daily temperature forcing further altered outbreak timing and severity (Figure 7). Transmission intensity varied seasonally, producing stronger outbreaks when host presence coincided with favourable temperatures. Small changes in transmission parameters produced nonlinear differences in epidemic size, highlighting the sensitivity of ATV dynamics to environmental conditions.

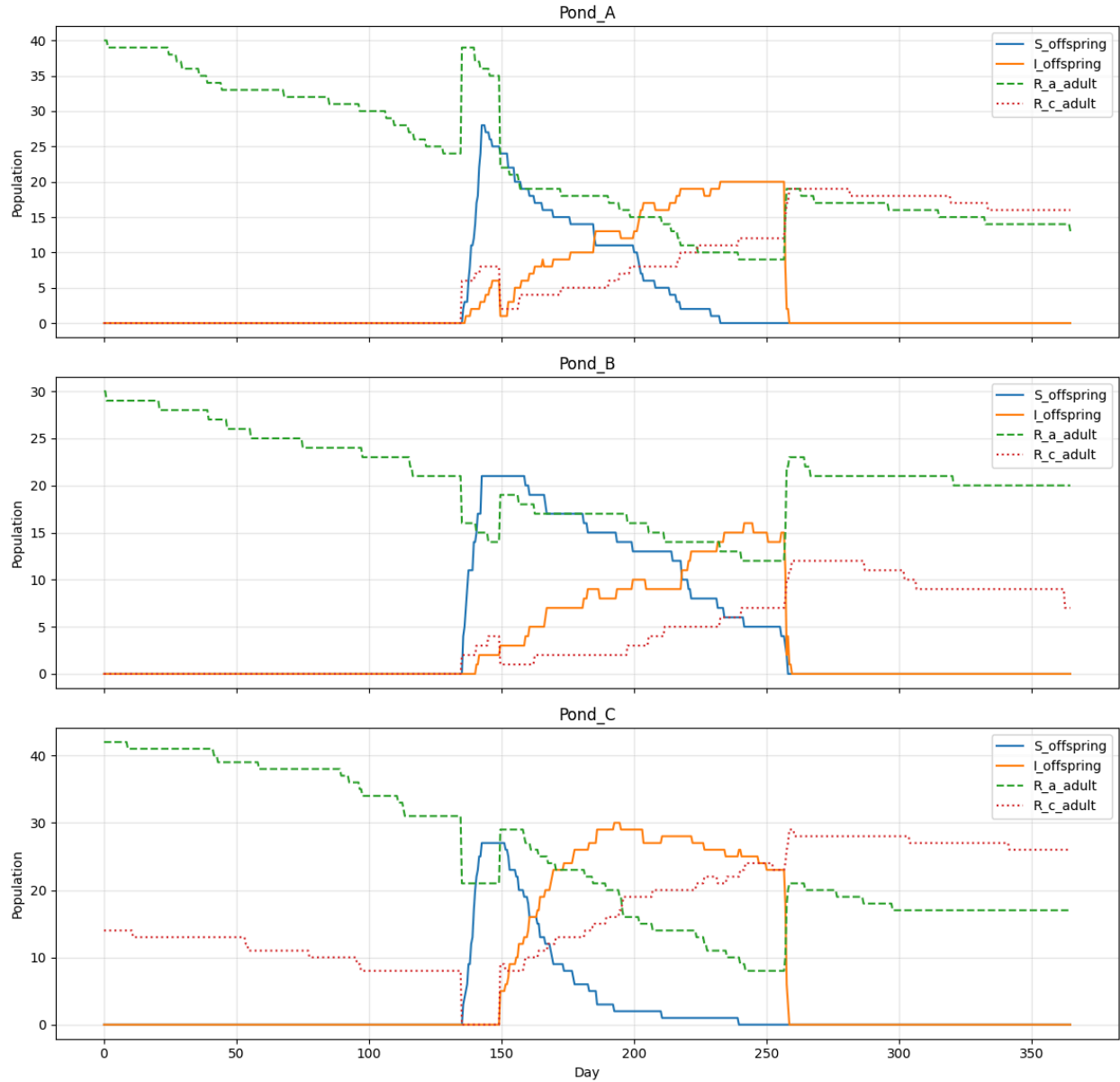


Figure 7. Model 3 output: effects of seasonal movement and temperature-dependent transmission on ATV dynamics. Lines show simulated compartment abundances through time when adult movement is combined with daily temperature forcing. Outbreak timing and magnitude depend on whether susceptible larvae, infected or carrier adults, and favourable transmission temperatures overlap during the breeding season.

Discussion

The observed ATV data showed clear variation across both years and regions. In 2019, cumulative infection prevalence increased earlier, rose more rapidly, and reached a higher final proportion than in 2020. Regional patterns also differed, with some regions showing stronger or earlier increases than others. These patterns suggest that ATV outbreaks are not simply determined by pathogen presence at individual ponds. Instead, outbreak severity likely depends on the timing of susceptible host availability, movement of infected or carrier adults, and environmental conditions that favour transmission. The progression from Models 1–3 supports this interpretation by showing that within-pond transmission alone cannot explain the temporal and spatial heterogeneity observed in ATV outbreaks. Rather, the interaction of host movement, breeding phenology, and climate forcing appears necessary to generate more realistic epidemic patterns. This supports the value of a mechanistic metapopulation framework for understanding ATV dynamics.

Previous ATV research has documented mortality events, local prevalence, and temperature-dependent susceptibility, but many studies have examined outbreaks at isolated sites (Brand et al., 2016; Brunner et al., 2007; Rojas et al., 2005). By linking within-pond transmission to between-pond movement and environmental forcing, the present model helps explain why outbreak timing and severity may differ among ponds, regions, and years.

Each model scenario helps clarify a different part of the outbreak process. Model 1 shows that within-pond transmission can generate local outbreaks, but without movement or environmental forcing, the system has no mechanism for pathogen spread among ponds or for environmentally driven differences in outbreak timing. Model 2 demonstrates that seasonal adult

movement can transform isolated pond outbreaks into a connected disease process, where infected or carrier adults introduce ATV into new breeding habitats and create secondary outbreaks. Model 3 adds an environmental filter by showing that movement alone is not enough to determine outbreak severity; transmission depends on whether susceptible larvae and infected or carrier adults are present during periods of favourable temperature. Together, these results suggest that severe ATV outbreaks are most likely when three conditions align: connected ponds allow pathogen movement, breeding produces susceptible larvae, and environmental conditions support efficient transmission.

These model outcomes are consistent with several biological mechanisms. Movement allows infected adults to introduce ATV into previously uninfected ponds and may synchronize outbreaks across connected habitats (Keeling & Rohani, 2008). Breeding phenology determines when susceptible larvae are present, while temperature may increase viral replication and reduce host immune performance (Price et al., 2018; Rollins-Smith & Le Sage, 2023). The strongest outbreaks, therefore, appear to occur when infected or carrier adults, susceptible larvae, and favourable environmental conditions overlap.

Several limitations remain. The model assumes homogeneous mixing within ponds, which likely simplifies real contact patterns because larvae and adults may vary in space use, density, and microhabitat within breeding sites. Movement and transmission parameters were approximate, so model outputs should be interpreted as mechanistic scenarios rather than calibrated predictions of specific outbreak sizes. The model also simplifies life-stage transitions by grouping individuals into broad offspring and adult classes, even though real salamander development, maturation, and survival may vary among ponds and years. Temperature effects were represented as a transmission modifier rather than as a fully parameterized physiological

response affecting immunity, viral replication, mortality, and host behaviour separately. In addition, the current movement structure remains a simplified representation of real salamander dispersal and does not yet account for pond-specific barriers, habitat quality, or variation in movement success. Future versions could address these limitations by calibrating parameters with field prevalence data, adding pond-specific environmental conditions, refining movement kernels, and incorporating an explicit environmental reservoir for viral persistence.

Building on these limitations, future work will focus on completing Model 4, which would incorporate pond-specific environmental conditions, heterogeneous temperature regimes, and broader Arizona landscapes. This would allow direct comparison of northern, central, and southern populations under distinct climate conditions. Additional improvements include parameter calibration using field prevalence data, refined movement kernels, and future climate scenarios to test whether warming shifts outbreak timing or increases epidemic severity. More broadly, this framework may help identify populations most vulnerable to ATV under future environmental change.

Conclusion

This study demonstrates that *Ambystoma tigrinum* virus dynamics in Arizona tiger salamander populations are best understood as the product of interacting ecological processes rather than as isolated within-pond outbreaks. While local transmission can generate events, the addition of seasonal host movement and temperature-dependent forcing produced more realistic variation in outbreak timing, magnitude, and spread across connected populations. These results highlight the importance of considering landscape connectivity, breeding phenology, and environmental variability when examining wildlife disease systems. Overall, this project shows

how spatially explicit mechanistic models can be used to bridge field observations and ecological theory, providing a framework for predicting disease risk under future climate and informing conservation efforts for vulnerable amphibian populations.

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Data Availability Statement

Scripts, model code, and supporting materials used to reproduce the analyses in this paper are available at: https://github.com/adk254/HURA_25-26